

Supplementary materials

Calculation of productivity loss

To account for income variations by age and sex, the gender pay gap per age group was applied as an alternative calculation of productivity loss due to CA-ARTI. The gender-pay gap (hereafter referred to as ‘gpg’) indicator as published by Eurostat represents the difference between average gross hourly earnings of male paid employees and of female paid employees as a percentage of average gross hourly earnings of male paid employees [127]. The gpg of Eurostat was given by year (2021-2023) and age group (ages =65). The gpg of 2023 was applied to 2024.

The Office for National Statistics (ONS) of the UK defines the gpg as follows: ‘the difference between average hourly earnings excluding overtime of men and women, as a proportion of men's average hourly earnings excluding overtime’ [128]. Although the age grouping of Eurostat was different, the ONS published the gpg by year (2021-2024, where estimates for 2024 are provisional) and age group as well (ages 18-21, 22-29, 30-39, 40-49, 50-59, 60+).

Geostat, the national statistics office of Georgia, defines the adjusted gpg as follows: ‘The ratio of the difference between the average salary for men and women to the average salary for men. ... The adjusted gender pay gap takes into account the demographics (age, education, marital status, place of residence – region/type of settlement, etc.) and job characteristics (field of economic activity, occupation, etc.)’ [143]. Gpg in Georgia was available for the years 2021 to 2023, but not specified per age group. The gpg of 2023 was applied to 2024.

Consequently, hourly labour cost for women and men per country, age group, and year were calculated as follows:

$$Labour\ cost\ women = \frac{hlc * 2}{(2 + gpg)}$$

$$Labour\ cost\ men = labour\ cost\ women * (1 + gpg)$$

where ‘hlc’ is the country-specific average hourly labour cost in Euros and ‘gpg’ is the gender pay gap. Productivity loss was then calculated by multiplying the hourly labour costs of men and women by the number of hours of work missed for each country, year, and age group. Results of the alternative calculation can be found in Table 1.

TABLE S1 | Hours of work missed, productivity loss in the base case and adjusted by sex and age group per country in Euros, and the percentual mutation between the base case and adjusted calculation.

Country	Mean work hours missed (no. patients)	Mean productivity loss (EUR), base case	Mean productivity loss (EUR), adjusted for sex and age group	Mutation of the adjusted calculation from base case (%)
Belgium	11.7 (62)	540.52	541.06	+0.1
Germany	30.6 (132)	1260.42	1228.77	-2.5
Spain	13.6 (46)	329.59	329.17	-0.1
France	7 (9)	290.56	282	-2.9
Georgia	8.3 (283)	24.04	23.45	-2.4
Greece	8.2 (179)	129.22	129.31	+0.1
Hungary	26.4 (144)	315.92	309.7	-2
Ireland	17.3 (119)	675.55	671.27	-0.6
Poland	26 (145)	380.77	376.74	-1.1
United Kingdom	7.7 (213)	227.4	225.98	-0.6
Average	15.4 (1332)	358.2	353.19	-1.4